Thermic Regrading under Norton Multiplication and the Newton-Polygon Boundary

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Abstract

Let $A_u(G) = G.u$ denote Norton multiplication of a short partizan game by a positive canonical dyadic $u = m/2^k$. Set $\delta_u = 2^{-k}$, with $\delta_u = 1$ when u is integral, and put $a_u = u - \delta_u$. For every nonnumeric short game,

$$\operatorname{mean}(A_u(G)) = u \operatorname{mean}(G), \qquad \operatorname{temp}(A_u(G)) = u \operatorname{temp}(G) + a_u.$$

For a number x of canonical mesh $\epsilon_x = 1/\operatorname{den}(x)$, the mean is ux and the temperature is -1 or $a_u - u\epsilon_x$, according as the latter is negative or nonnegative. Every positive numeric Norton operator therefore induces additive maps between thermic residue groups. These maps do not form an action of the multiplicative dyadics: the exact degree defect is

$$\Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0,$$

and we classify its zero set. Thus thermography has a precise tropical shadow, but not a Newton-polygon-style graded ring: freezing is not a tropical operation, nonnumeric Norton units fail to descend modulo colder games, and the temperature-zero residue contains nonzero 2-torsion.

1 Introduction

Temperature measures the cooling required before a short partizan game freezes to a number. It therefore filters the additive group of short games. Norton multiplication $G.U$ is additive in G for each positive game U [4, Chapter 8]; further accounts and applications appear in [7, 8]. We determine the induced map on the thermic filtration when U is a positive number.

The degree law is affine rather than homogeneous. For a positive canonical dyadic u and a nonnumeric game G, it is

$$\tau \longmapsto r_u(\tau) = u\tau + u - \delta_u.$$

The denominator mesh δ_u is essential and produces the composition defect $\Delta(u, v)$. Sections 2 and 3 derive the complete wall transform, including the separate numeric branch. Sections 4 and 5 pass to thermic residue groups and determine exactly when successive transports have the same degree as multiplication by uv.

Section 6 compares this structure with Newton polygons. Both theories use filtrations and extremal piecewise-linear envelopes, but thermographic freezing has no tropical analogue and the thermograph is not a congruence for disjunctive sum. The comparison is therefore a precise limitation, not an identification [3, 6].

2 Norton multiplication and thermography

We work with short normal-play partizan games modulo Conway equality. Numbers have temperature -1. For every nonnumber G, its left and right thermographic walls meet at a finite height

$\operatorname{temp}(G) \geq 0$ and then freeze to the common mast $\operatorname{mean}(G)$. We use the standard thermographic conventions of [10, 3].

Norton multiplication depends on the selected form of its positive unit, so throughout a numeric unit is taken in its canonical short-game form. Write a positive dyadic in lowest terms as $u = m/2^k$ and set

$$\delta_u = \frac{1}{\operatorname{den}(u)} = 2^{-k}, \qquad a_u = u - \delta_u. \tag{1}$$

For an integer u, this gives $\delta_u = 1$. A nonintegral canonical dyadic has options $u - \delta_u$ and $u + \delta_u$, while a positive integer has the sole left option $u - 1$. Every Norton incentive is consequently a_u, and the defining recursion becomes

$$A_u(n) = nu \quad (n \in \mathbb{Z}), \qquad A_u(G) = \{A_u(G^L) + a_u \mid A_u(G^R) - a_u\} \quad (G \notin \mathbb{Z}). \tag{2}$$

For a number x, put $LS_t(x) = RS_t(x) = x$. For a nonnumber G, let

$$L_G^{\mathrm{raw}}(t) = \max_{G^L} RS_t(G^L), \qquad R_G^{\mathrm{raw}}(t) = \min_{G^R} LS_t(G^R), \tag{3}$$

$$LS_t^{\mathrm{sc}}(G) = L_G^{\mathrm{raw}}(t) - t, \qquad RS_t^{\mathrm{sc}}(G) = R_G^{\mathrm{raw}}(t) + t. \tag{4}$$

The scaffold walls are followed until their first meeting, after which both are replaced by the common constant mast. The raw gap is

$$D_G(t) = L_G^{\mathrm{raw}}(t) - R_G^{\mathrm{raw}}(t) - 2t. \tag{5}$$

It is nonincreasing, and $\operatorname{temp}(G)$ is its first zero. We shall also use the standard facts that left walls are nonincreasing, right walls are nondecreasing, and wall slopes lie between -1 and 1.

3 The thermic transformation law

The recursion stops at integers, not at all numbers. Numeric inputs therefore form a separate branch of the theorem.

Lemma 3.1 (Numeric inputs)**.** *Let x be a short-game number and put $\epsilon_x = 1/\operatorname{den}(x)$. Then*

$$\operatorname{mean}(A_u(x)) = ux, \qquad \operatorname{temp}(A_u(x)) = \begin{cases} -1, & a_u - u\epsilon_x < 0, \\ a_u - u\epsilon_x, & a_u - u\epsilon_x \geq 0. \end{cases}$$

In particular, $\operatorname{temp}(A_u(x)) < a_u$.

Proof. Induct on the exponent of the exact dyadic denominator of x. Integers are the base case of the Norton recursion. If x is nonintegral, then $x = \{x - \epsilon \mid x + \epsilon\}$ with $\epsilon = \epsilon_x$. The option images have means $u(x - \epsilon)$ and $u(x + \epsilon)$ and, by induction, freeze strictly below $a_u - u\epsilon$ whenever that value is nonnegative. At $c = a_u - u\epsilon \geq 0$, the parent scaffold values are

$$u(x - \epsilon) + a_u - c = ux, \qquad u(x + \epsilon) - a_u + c = ux.$$

Immediately below c their gap is $2(c - t) > 0$, so the first meeting is at c and has mast ux. If $c < 0$, the two numeric bounds in the Norton recursion lie on opposite sides of ux; the simplicity rule selects the number ux. $\square$

The central wall identity is affine covariance of the raw gap.

Lemma 3.2 (Raw-gap transport)**.** *Suppose the wall formula*

$$LS_{a_u+ut}(A_u(X)) = uLS_t(X), \qquad RS_{a_u+ut}(A_u(X)) = uRS_t(X)$$

holds for every option X of a nonintegral game G. Then

$$L^{\mathrm{raw}}_{A_u(G)}(a_u + ut) = uL^{\mathrm{raw}}_G(t) + a_u, \tag{6}$$

$$R^{\mathrm{raw}}_{A_u(G)}(a_u + ut) = uR^{\mathrm{raw}}_G(t) - a_u, \tag{7}$$

and therefore

$$D_{A_u(G)}(a_u + ut) = uD_G(t). \tag{8}$$

Proof. A left option of $A_u(G)$ is $A_u(G^L) + a_u$, and numeric translation shifts both walls by a_u. Taking the maximum in the definition of the raw left wall proves its displayed transform; the right-option minimum gives the raw right transform. Substitution into the raw-gap definition, with $s = a_u + ut$, cancels the two parent shifts and yields the displayed covariance. □

When $\mathrm{temp}(G) > 0$, raw-gap covariance immediately locates the first meeting. The only subtle case is $\mathrm{temp}(G) = 0 < a_u$: one must exclude a premature mast below a_u. We isolate the strategy argument needed for that case.

Lemma 3.3 (No premature mast)**.** *Let X be a canonical nonnumber and m a number. Assume the wall formula of Lemma 3.2 for every proper follower of X. If, for some $b < a_u$,*

$$RS_s(A_u(X)) = um - a_u + s \qquad (b \le s \le a_u),$$

then Right wins $X - m$ moving first. Dually, if $LS_s(A_u(X)) = um + a_u - s$ on the same interval, then Left wins $X - m$ moving first.

Proof. We prove the right-wall statement by birthday induction; negation gives the dual statement. The displayed wall has maximal slope 1, so it is an unfrozen right scaffold on $[b, a_u)$. By the Norton recursion,

$$\min_{Y \in X^R} LS_s(A_u(Y)) = um \qquad (b \le s \le a_u).$$

Choose a right option Y attaining this minimum at b. Left walls are nonincreasing and every candidate is at least um throughout the interval, so $LS_s(A_u(Y)) = um$ on $[b, a_u]$.

If Y is numeric, Lemma 3.1 shows that it has frozen below a_u with mast uY, hence $Y = m$. Otherwise, for every left option Z of Y, the parent scaffold inequality gives

$$RS_s(A_u(Z)) + a_u - s \le um.$$

At $s = a_u$, the induction hypothesis implies $RS(Z) \le m$. If the inequality is strict, the standard stop criterion says that Right wins $Z - m$ moving first. If equality holds, the corridor equality excludes numeric Z by Lemma 3.1: a numeric image is frozen strictly below a_u, whereas this wall has slope one on a nontrivial interval. The slope bound therefore forces equality throughout the corridor, and birthday induction gives the same conclusion. Right moves from $X - m$ to $Y - m$ and answers every left reply using this strategy. Number avoidance shows that a move in the numeric component is no better [10, Part II]. □

Theorem 3.4 (Thermic regrading). *For every positive canonical dyadic u and every nonnumeric short game G,*

$$\operatorname{mean}(A_u(G)) = u \operatorname{mean}(G), \qquad \operatorname{temp}(A_u(G)) = u \operatorname{temp}(G) + a_u. \tag{9}$$

More strongly, for every $t \geq 0$,

$$LS_{a_u+ut}(A_u(G)) = uLS_t(G), \qquad RS_{a_u+ut}(A_u(G)) = uRS_t(G). \tag{10}$$

For numeric G, Lemma 3.1 is the exact replacement.

Proof. Proceed by birthday induction, using Lemma 3.1 for numeric followers. Lemma 3.2 then applies to G. Put $\tau = \operatorname{temp}(G)$. If $\tau > 0$, then $D_{A_u(G)}(a_u) = uD_G(0) > 0$, so monotonicity excludes an earlier meeting; raw-gap covariance places the first zero at $a_u + u\tau$. If $\tau = 0 = a_u$, the same identity places it at zero.

Suppose $\tau = 0 < a_u$, and write $LS(G) = RS(G) = m$. Raw-gap covariance makes the transformed scaffolds meet at (a_u, um). If $A_u(G)$ had frozen at some $b < a_u$, its walls would equal um on $[b, a_u]$. For each left option X of G, either $RS(X) < m$, in which case Right wins $X - m$ by the stop criterion, or the slope bound forces

$$RS_s(A_u(X)) = um - a_u + s$$

on the corridor, and Lemma 3.3 gives the same conclusion. Thus Right wins $G - m$ playing second. The dual argument for every right option shows that Left also wins $G - m$ playing second, whence $G = m$, contrary to the assumption that G is nonnumeric. Premature freezing is impossible.

Before the first meeting, the raw left-wall transform and the scaffold definition give

$$LS^{\mathrm{sc}}_{a_u+ut}(A_u(G)) = uL^{\mathrm{raw}}_G(t) + a_u - (a_u + ut) = uLS^{\mathrm{sc}}_t(G),$$

and similarly on the right. At and above the meeting both sides freeze to the mast $u \operatorname{mean}(G)$. This proves both assertions. □

4 Thermic residue transport

Let $F_{<0} = \mathcal{N}$ be the subgroup of short numbers. For $\tau \geq 0$, let $F_{\leq\tau}$ be the subgroup of games of temperature at most τ, including $F_{<0}$, and let $F_{<\tau}$ be the union of lower layers. The standard inequality

$$\operatorname{temp}(G + H) \leq \max\{\operatorname{temp}(G), \operatorname{temp}(H)\} \tag{11}$$

shows that this is an additive filtration [10, Theorem II.5.18].

Corollary 4.1. *For every $\tau \geq 0$, Norton multiplication by u induces an additive map*

$$\overline{A}_{u,\tau} : F_{\leq\tau}/F_{<\tau} \longrightarrow F_{\leq u\tau+a_u}/F_{<u\tau+a_u}.$$

For a positive integer n, the shifted thermal height $h(G) = \operatorname{temp}(G) + 1$ satisfies $h(A_n(G)) = nh(G)$ on nonnumeric layers.

Proof. Norton multiplication by a fixed positive unit is additive [4, 8]. If $G - H$ has temperature $\sigma < \tau$, Theorem 3.4 sends it to temperature $u\sigma + a_u < u\tau + a_u$. If $G - H$ is numeric, Lemma 3.1 puts its image strictly below a_u. Hence the map is well defined on the quotient. For integral n, $a_n = n - 1$. □

5 The composition defect

Write $r_u(\tau) = u\tau + u - \delta_u$. Successive thermic transports differ from transport by the ordinary dyadic product by a degree-independent term.

Theorem 5.1. *For positive dyadics u, v and every $\tau \geq 0$,*

$$r_v(r_u(\tau)) = r_{uv}(\tau) + \Delta(u,v), \qquad \Delta(u,v) = v(1-\delta_u) - \delta_v + \delta_{uv} \geq 0. \tag{12}$$

The defect is zero exactly in the following cases:

1. *u is nonintegral and $v = 2^{-\ell}$ for some $\ell \geq 0$;*
2. *u is an odd positive integer and v is arbitrary;*
3. *u is an even positive integer and v is an integer.*

Proof. Expanding the affine maps gives

$$r_v(r_u(\tau)) - r_{uv}(\tau) = v(1-\delta_u) - \delta_v + \delta_{uv}.$$

If u is integral, then $\delta_u = 1$ and $\Delta = \delta_{uv} - \delta_v$. Multiplication by an integer cannot increase a dyadic denominator. It preserves every denominator when u is odd; for even u, it preserves the denominator exactly when v is integral. This proves the integral cases and their nonnegativity.

Now let $u = m/2^k$ in lowest terms with $k > 0$. If v is nonintegral, then $\delta_{uv} = \delta_u \delta_v$, and

$$\Delta = (1-\delta_u)(v - \delta_v) \geq 0.$$

Equality holds precisely when the numerator of v is 1. If $v = n$ is an integer, write $n = 2^s q$ with q odd. For $s \geq k$, $\delta_{uv} = 1$ and the defect is positive unless $n = 1$. For $s < k$,

$$\Delta = n - 1 - \frac{n - 2^s}{2^k},$$

which is zero at $n = 1$ and positive for $n > 1$. This completes the classification. □

Corollary 5.2. *The residue transports of Corollary 4.1 cannot be the scalar maps of an associative action whose positive dyadic coefficients multiply ordinarily and whose equivalence preserves thermic degree.*

Proof. Take $u = 1/2$, $v = 2$, and $G = *$. Theorem 5.1 gives $\Delta(1/2, 2) = 1$, so

$$\mathrm{temp}(A_2(A_{1/2}(*))) = 1, \qquad \mathrm{temp}(A_1(*)) = 0.$$

Associativity would identify these two classes, but an equivalence preserving thermic degree cannot do so. □

6 Comparison with Newton polygons

The filtration inequality above is analogous to the nonarchimedean inequality, and the raw option operators are pointwise max-plus and min-plus folds. Cooling adds the affine terms $-t$ and $+t$. This is the exact tropical content of thermography.

The subsequent freeze is different: one compares the two envelopes, finds the first zero of D_G, and replaces both walls by one constant mast. Nor is the thermograph a congruence for disjunctive sum. The games $* = \{0 \mid 0\}$ and $\uparrow = \{0 \mid *\}$ have the same constant-zero walls, but $* + * = 0$ is numeric while $\uparrow + *$ is a nonnumber of temperature zero. Thus no binary operation on thermographs alone recovers the thermograph of every sum.

Newton polygons, by contrast, arise from valuations of coefficients and transform functorially under polynomial multiplication; Dumas's theorem is a classical form of this product behavior [6]. The game side has an additive filtration and the external transports of Corollary 4.1, but no analogous internal product on all Conway game values.

The restriction to numeric Norton units is also necessary. At degree zero, $*$ and $*+1$ differ by the cold number 1. For the positive nonnumeric unit $U = \uparrow$, additivity gives

$$A_U(*+1) - A_U(*) = A_U(1) = \uparrow,$$

whose temperature-zero class is nonzero. Thus A_U does not descend modulo cold numbers. Finding coarser equivalences compatible with Norton multiplication is a separate problem of the kind posed by [2, Problem 6.10].

Finally,

$$\mathrm{gr}_0(\mathcal{G}) = F_{\le 0}/F_{<0} \cong \mathcal{I},$$

the additive group of infinitesimals [5, 9]. It contains $[*] \neq 0$ with $2[*] = 0$.

Proposition 6.1 (Dyadic torsion obstruction)**.** *There is no faithful module containing* $\mathrm{gr}_0(\mathcal{G})$ *for which multiplication by a coefficient* 2 *agrees with game doubling and* 2 *has a multiplicative inverse representing* $1/2$*.*

Proof. Let s be the image of $[*]$. Compatibility with game doubling gives $2s = 0$. If 2 is invertible, then $s = (1/2)(2s) = 0$, contradicting faithfulness. □

This obstruction does not exclude integer coefficients, quotients that kill the incompatible torsion, or finer game equivalences. It excludes the specific faithful full-dyadic graded-ring reading.

7 Computational realization

Ogdoad [1] implements recursive Norton multiplication and overheating together with closed-form routines for numeric regrading, mean–temperature transport, and the composition defect. Exact tests compare these formulas with recursively materialized products on bounded families of short games and canonical dyadic units. A separate transcription of the Norton recursion checks the nonintegral-unit branches. These computations are regression oracles for the proved formulas, not finite substitutes for the universal argument.

The implementation exposes the max-plus/min-plus option folds separately from the ordinary meeting-and-freezing step. It consequently realizes the same structural boundary as Section 6.

8 Conclusion

Positive numeric Norton multiplication transforms every nonnumeric thermic degree by the affine map

$$\tau \longmapsto u\tau + u - \delta_u,$$

with a separate, strictly lower branch for numeric inputs. The resulting maps on thermic residues are additive, and the nonnegative defect $\Delta(u, v)$ measures their exact failure to form a multiplicative dyadic action. Thermography therefore supports a filtered tropical analogy through its extremal envelopes and nonarchimedean temperature inequality, but freezing, nonnumeric Norton operators, and temperature-zero torsion rule out a faithful Newton-style graded ring.

References

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